BAT
M6 terminal, BMS P+
M4 terminal, BMS P−
LM5069MMX-1
IPB017N10N5LFATMA1
0.5mR CSS4J-4026R-L500F 5W
MMBTA92
TPS7A4001DGNR
TP VCTL
E-stop: 1 NC out, 2 GND, 3 NC return, 4 GND, 5−6 reset NO, 7 GND, 8 button NO
MMBT3906
MMBT3904
TP RUN
TLV1805DBVR
LM4040AIM3-2.5
BSC040N10NS5ATMA1
M4 terminal, HS50 +
M4 terminal, HS50 −
12 V converter pigtail: 1−2 +VIN, 3−4 −VIN, 5 ON/OFF (low = on), 6 −VIN, 7−9 −VOUT, 10−12 +VOUT
12 V to Jetson
12 V to Pi 5 V buck
12 V to fans
Pi: 3V3, SDA, SCL, GND, E-stop pressed (low)
INA228AIDGSR
2N7002
LM5069MMX-2
1mR CSS4J-4026R-1L00F 6W
L arm +
R arm +
waist +
aux +
L leg +
R leg +
GND stud 1: L arm, R arm, waist returns
GND stud 2: aux, L leg, R leg returns
TPS48100QDGXRQ1
TPS25954ADSG
TP MID
TP VBUS
SPARE_SW, off with the motors
SPARE_ON, always on

## 1. Battery in and E−stop bank B (LM5069−1, two FETs). Bank A, in series ahead of it, is block 7

The bank is the soft start for all the motor capacitance: BMS sees no inrush. Reset closes it, loop open drops it. No software in the path. Compute is not behind it.
Rs 0.5 mOhm: limit 97−123 A (VCL 48.5−61.5 mV), breaker 160−260 A (VCB 80−130 mV). Above peak draw and JK DOC 80 A. 1.8 W at 60 A.
Start: Cdv 100 nF, gate 10−22 uA, 100−220 V/s, 0.25−0.55 s to full. Bus ceiling 9.4 mF at 2.06 A hot DC SOA line (112 W, 45 K Tj rise). Drive caps unmeasured.
Power limit 652 W at 54.6 V, 480−810 W with tolerance: RPWR = 1.3e5 x Rs x (P − 1.18 mV x VDS / Rs), SNVS452G eq 10, VSNS 6 mV over the 5 mV floor. Timer 4.7 nF: 0.22 ms typ, 0.15−0.39 ms.
Fault check: 810 W is 14.8 A at 54.6 V for up to 0.39 ms. Reviewers read about 16 A at 0.82 ms, 70 C, one FET; TI scales as t^0.346, so about 20.7 A at 0.39 ms: 1.4 x. Read the real curve at layout.
FETs: 0.85 mOhm for the pair, 5.2 W hot at 60 A, 1.3 W at 30 A, the same again in bank A. UVLO low sinks the gate at 2 mA: about 170 us off.
The 2 mA also pulls VBUS down through U1's OUT−GATE zener, against the 54 uA of its internal 1 MOhm SENSE−OUT path, so VBUS reads near 0 V with the banks open.
Body diodes of both banks pass regen back to the pack with the banks open. OVLO tied low: opening on overvoltage would strand regen on the bus.
LM5069−1 latches on a fault. Clear: E−stop cycle, then reset.
BAT spike: the absorber in block 10 takes the lead current when a bank opens fast (breaker trip 1.2 us), 5.0SMDJ58A is the backstop. FETs and U1 are 100 V.

Block 10. Absorber: a bank trip into a bus short cuts 350−480 A. tools/absorber_sim.py (pack 54.6 V, 0.1 Ohm loop, 0.5−1 uH assumed, no pack yet) gives BAT 81−83 V peak with this 47 uF, 50 mOhm part and D1. Measure the lead at bring−up. 100k bleeds it. BMS close drives 190−230 A into it for a few us: bench it on the JK.
F1, 1 uF, BAV19W and SMAJ15A (M14, M6, M7): a shorted U2 puts the pack on VCTL; the TVS clamps under the 40 V parts and F1 opens. The BAV19W keeps C5 from back−driving U2 if BAT collapses.
Absorber close option (M3), DNP: if the JK trips when it closes onto C38, remove the 0R and fit 2R + the delayed NMOS + its RC: 25 A at close, clamp unchanged (absorber_sim.py).
RUN_A and RUN_B diodes: a U1 pin 2−3 bridge or an EN fault can't drive RUN. FLT reaches UVLO through 10k (2010, 0.75 W) with a 15 V zener on FLT_A: FLT low holds UVLO near 1.7 V, and a bridged UVLO puts at most 5.5 mA into FLT and 15 V on it (20 V max).
Converter: F2 20 A on the board end protects the pigtail (U7 already limits at 9.7−12.3 A; F2 backs up a shorted Q7).
The DVE has no reverse protection, so the keyed connector carries polarity. U7 PGD high (good) turns on the 2N7002, which pulls ON/OFF to −Vin: on. 470k and the 12 V zener bias the gate from BRK_IN, so it doesn't depend on VCTL.
1 uF on RST_SRC (M8): a reset held at power−up no longer arms the latch (tools/latch_sim.py, 36 cases). Reset interlock (M1): Q10 shorts RST_SRC until MID is over about 11 V through the 10 V zener and Q11, so with bank B shorted and the bus down, reset does nothing; healthy MID is 17 V at a 36 V pack. It loads MID by about 30 uA. 33k on U17 VBUS (M13). 22 nF SRC to GND, TI's recommendation for U8. BAT46W (M4). 330 uF on +12V: spare−output shorts and servo back−feed (M9) land in it and the compute loads.

## 2. Control rail and E−stop latch (no logic IC)

12.1 V, 11 mA. On the battery side, so regen through the body diodes keeps it up with the BMS open. Runs the latch and the clamp. 0.5 W at 54.6 V.
Fed through F1 with its own 1 uF, clamped by a 15 V TVS, reverse diode OUT to IN: block 10.

Seal−in latch, ngspice checked at hFE 80: stays off at power−up even with reset held, reset closes it, a 50 us contact bounce rides through, loop open drops it in about 100 us, re−closing the loop does nothing, a reset held through a loop cycle does nothing, a new press works 5 ms after release.
RUN 7 V. UVLO and EN_A each take RUN through their own diode (block 10), so a fault on either pin cannot drive RUN. UVLO 3.8 V; needs about 4.8 V on RUN_B rising. Loses VCTL, RUN drops, UVLO drops: bank opens.
Loop 6 mA, wetting for a control−rated NC contact. No 12 V on the connector without a series R. GND between loop out and return, and between return and reset: a short there opens the loop or does nothing.
Brake resistor NC thermostat: wire it in series in the loop harness, no board parts.
Button HB2−BS545, 22 mm mushroom, twist release, 1NC + 1NO, 660 V AC 10 A, control−rated only: NC in this loop, NO to the Pi (block 9). RUN also enables bank A (block 7) and SPARE_SW (block 8).

## 3. Braking clamp: 56.8 V on, 55.8 V off, 57.5 V worst case, into the HS50

Arcol HS50 10R J on the +X gusset plate (frame 3877317), leads to J3/J4 on the board's top edge, +X half, twisted: 322 W, 5.7 A while on. MBRS3100 carries the lead and winding inductance at turn−off, so lead length sets EMI, not FET stress. 50 W on Arcol's standard heatsink, 32 g: [illegible]
Self−powered: VCTL lives as long as the bus or the pack does. If the FET sticks on, the plate thermostat opens the loop and the bank drops the bus.
TLV1805 drives the FET directly: 100 mA peak, 0.3 us into 4 nF. Rev A's driver, meter, shunt, status FET and NTC are gone, and a smaller FET replaces the IPT015.
Leg stage later: fit 2 Ohm and recheck this FET, 29 A gives about 5 W.

## 4. Bus capacitors

400 uF, 80 V. The drives have their own. Can is 10 x 16.5: land owed, and it only clears under the higher ceiling, away from the fans.

## 5. 12 V converter: Murata DVE0125V2NBC on the −X front post plate, pigtail J24 (block 10)

Ahead of the E−stop: motors off, compute stays up. Soft start in block 5a; the converter's negative−logic ON/OFF is held off until U7's PGD is good (block 10).
Off board on the top front rail (frame 3877317), baseplate down, 4x M3 into its threaded baseplate (50.8 x 15.24 mm, 3 mm max depth). J24 sits on the board's top edge, −X half. −Vin and −Vout both go to GND: the robot has one ground, P−.
Pigtail about 150 mm, twisted pairs, 18 AWG: 2 wires each way on the input, 3 each way on +12 V (0.1 V drop at 12 A including contacts). At the converter end: 33 uF 100 V aluminium (its ESR damps the lead, 75−150 nH) plus 1 uF across +VIN/−VIN at the pins. 300 W, 25 A, 94.6 %: about 9 W at the 140 W ceiling, into the plate.
Input UVLO 32−36 V on, 30−34 V off, fixed on the DVE: near an empty pack under sag, compute can drop before the BMS cuts out.
INA228 addresses: L arm 0x40, R arm 0x41, waist 0x42, aux 0x43, L leg 0x44, R leg 0x45, 12 V 0x46, SPARE_SW 0x47, MID 0x48. Powered by the Pi's 3.3 V; Pi pull−ups on I2C.

## 5a. Brick soft start (ahead of the E−stop switch)

Same controller as the bank, so one part family. −2 retries on its own: a compute fault does not need a BMS cycle. UVLO on VIN, OVLO low: on whenever the pack is.
Power−limit start, no dV/dt network: 70 W limit (Rs 5 mOhm, same equation as U1), about 1.3 A at full pack. 4.4 uF plus the brick's own input caps (not published) charges in under 1 ms at 20 uF.
Timer 100 nF: 4.7 ms typ, 3.1−8 ms. Limit 9.7−12.3 A clears the converter's worst draw, about 4.2 A at 36 V. It charges about 48 uF at the converter end plus the DVE's unpublished internal capacitance at the low power limit.
Fault check: 87 W is 1.6 A for up to 3.8 ms, against 12 A at 10 ms on the datasheet SOA. Q7 is the linear FET for that reason.
Compute is on whenever the BMS is on. Loss at 3 A: 0.05 W shunt, 0.03 W FET.

## 10. Rev B2 protection parts (refs appended so earlier refs keep their numbers)

## 8a. SPARE_ON eFuse

Unswitched, about 2 A limit. Same part as U9: a short stops in 5 us and leaves compute up.

## 10b. Absorber close option (M3): 0R fitted, the rest DNP

## 9. E−stop pressed: button NO contact to the Pi

Low means pressed. Pi GPIO input, no Pi pull−up needed. 3.3 V through 3.3k gives the contact 1 mA of wetting.
A harness short to the 54.6 V bus: the BAT46W blocks the pull−up (block 10), and 100k leaves 0.5 mA in the clamp pair. Open contact reads about 3.0 V.
Information only: the NC loop does the stopping.

## 6. Limb outputs: terminal pair, 1 mOhm shunt, INA228. Fuses inline in the pigtails.

[illegible] values are rev A's, unchanged.
Legs DNP: fit the same parts when they arrive.
Returns share two GND studs, three limbs each: check the stud and its lug for the summed current.

## 7. E−stop bank A: second series bank, own driver, ahead of bank B

Own driver: a FET usually dies shorted gate to drain, which would pull a shared gate net to the pack and hold the other bank on. RUN is the only shared signal, and each bank takes it through its own diode.
Both banks face the same way, drains toward the pack, so regen stays open. Bank B keeps U1, the shunt and the soft start.
Close order: U8 FLT holds U1 UVLO low through a 100 V diode until BST is up (about 57 ms typ, 35−115 ms with 2.2 uF). The diode keeps a UVLO fault off FLT (20 V max), so A closes first by logic, not by timing. A closes onto MID only. FLT also blocks B if U8 has no charge pump or has tripped.
Off edge: EN falls first, 4.6 us typ, 6.5 us max, then G1PD pulls 390 nC through 22R: about 2−3 us. The absorber (block 10) takes the lead's current, so a slow edge buys nothing.
Short−circuit comparator on RS1: 40.2k gives 99 mV, 134−313 A with the ISCP spread, above U1's limit. TMR 100k only (C29 DNP): fastest response, 4.3 us, still latch−off, cleared by an E−stop cycle. It covers reset with B shorted: A opens, and FLT holds B off.
Driver off holds gate to source with no path to GND, so MID can be biased: 47k + 47k, 0.48−0.50 x pack, 0.6 mA, 63 mW each. Hot FET leakage, up to 200 uA per bank, moves MID by under 5 V. U17 reads MID through 33k (spike and fault limit): calibrate the 3 % gain.
Self−test before reset, pack 36−54.6 V, BMS telemetry for pack voltage: MID 0.4−0.6 x pack and U11 VBUS under 3 V is healthy. MID within 2 V of pack: A shorted. MID under 3 V: B shorted. VBUS high: both shorted, or the bus has not bled down yet, about 11 s on board caps, longer with drives.
U17 uses VBUS only, IN+ and IN− on GND. VBUS alone cannot see one shorted bank, which is why MID is read.

## 8. Spare 12 V outputs

SPARE_SW: RUN enables it, so it drops with the motor bank. The divider keeps 7.7 V of RUN under the 6 V EN limit. Limit 3.87−4.42 A (487R). dV/dt 3.3 nF: about 12 V/ms. One switch: a shorted U9 stays on. Current and volts at 0x47, 5 mOhm, ADCRANGE 1.
SPARE_ON: block 8a. Both spares sit behind RS8, so the 12 V meter counts them.

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/electrical/pdb/REV_B_PAR
**marengo**
Sheet: /
File: marengo_pdu_revb.kicad_sch
***Title: marengo power board rev B***
Size: A1 | Date: 2026−11−06 | **Rev: B−wip**
KiCad E.D.A. 10.0.8 | Id: 1/1